



RTL Design Sherpa

APB PM/ACPI Micro-Architecture Specification 1.0

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1 Pm Acpi Mas Index

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2 APB PM/ACPI - Overview

2.1 Introduction

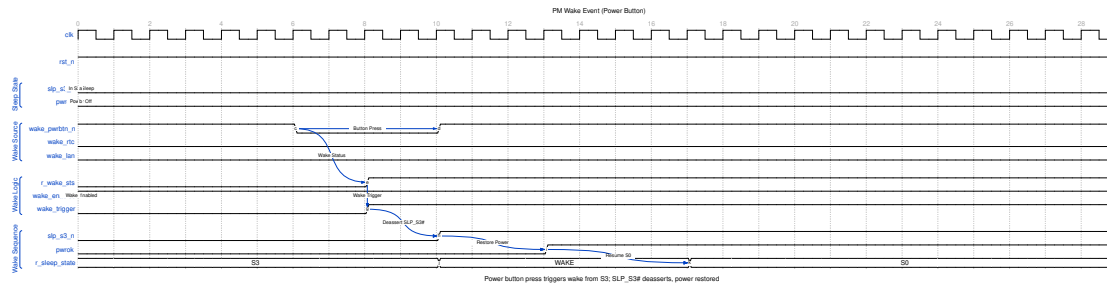
The APB PM/ACPI controller provides ACPI-compatible power management functionality with an APB interface. It handles system power states, events, and timer functionality.

2.2 Key Features

- ACPI-style power management events
- Single PM1 control/status/enable block (no separate PM1a/PM1b)
- PM timer (32-bit, 3.579545 MHz equivalent via configurable divider)
- GPE (General Purpose Events) support: 32 sources in one bank
- Clock gating control (32 domains) and power domain control (8 domains)
- System sleep state control (S0/S1/S3)
- Single active-high pm_interrupt output (no separate SCI/SMI outputs)

2.3 Applications

- System power management
- Sleep state transitions (S0/S1/S3)
- Wake event handling
- Power button events

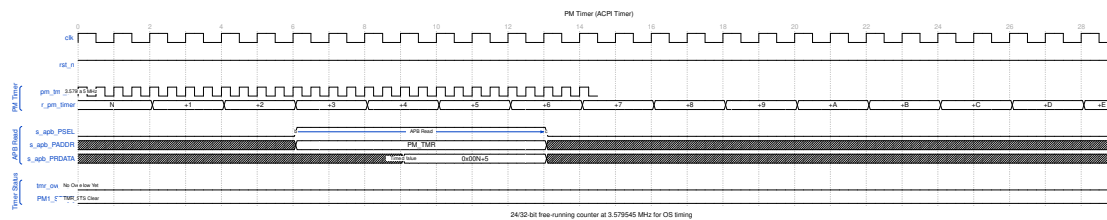


PM Wake Event

Wake sequence: 1. Enabled wake source detected (power button, RTC alarm, external, or GPE) 2. Wake status latched in WAKE_STATUS / PM1_STATUS.wak_sts 3. The FSM transitions back to S0 4. pm_interrupt asserts if the corresponding enable is set

2.5.3 Waveform 1.3: PM Timer

Free-running PM timer for timing services.

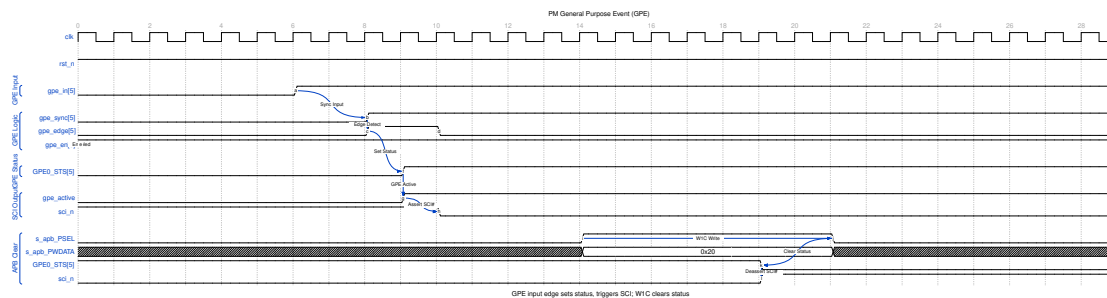


PM Timer

The 32-bit free-running counter (PM_TIMER_VALUE, 0x020) increments at a 3.579545 MHz equivalent rate using the PM_TIMER_CONFIG divider (default 0x001B, divide-by-28). Overflow sets the timer_overflow / tmr_sts status.

2.5.4 Waveform 1.4: General Purpose Event (GPE)

External events set a GPE status bit and can raise pm_interrupt.



PM GPE Event

A GPE input edge sets a bit in GPE0_STATUS_LO/HI. If the matching GPE0_ENABLE_LO/HI bit is set, the aggregated GPE interrupt asserts pm_interrupt. Software reads status, services the event, then writes 1-to-clear the status bit.

2.6 Register Summary

Selected registers; see [Chapter 5](#) for the complete map, fields, resets, and access types.

Offset	Name	Access	Description
0x000	ACPI_CTRL	RW	Global control and power state
0x004	ACPI_STATUS	W1C	Global status and power events
0x008	ACPI_INT_ENABLE	RW	Interrupt enable mask
0x00C	ACPI_INT_STATUS	W1C	Interrupt status
0x010	PM1_CONTROL	RW	PM1 control (sleep, button override)
0x014	PM1_STATUS	W1C	PM1 status flags
0x018	PM1_ENABLE	RW	PM1 event enable mask
0x020	PM_TIMER_VALUE	RO	PM Timer current value (32-bit)
0x024	PM_TIMER_CONFIG	RW	PM Timer clock divider
0x030	GPE0_STATUS_LO	W1C	GPE status bits [15:0]
0x034	GPE0_STATUS_HI	W1C	GPE status bits [31:16]
0x038	GPE0_ENABLE	RW	GPE enable

Offset	Name	Access	Description
	_LO		bits [15:0]
0x03C	GPE0_ENABLE	RW	GPE enable
	_HI		bits [31:16]
0x050	CLOCK_GATE_CTRL	RW	Clock gating control [31:0]
0x054	CLOCK_GATE_STATUS	RO	Clock gate status
0x058	POWER_DOMAIN_CTRL	RW	Power domain control [7:0]
0x05C	POWER_DOMAIN_STATUS	RO	Power domain status
0x060	WAKE_STATUS	W1C	Wake event sources
0x064	WAKE_ENABLE	RW	Wake event enable mask
0x068	RESET_CTRL	RW	Reset generation control
0x06C	RESET_STATUS	RO	Reset source information

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3 APB PM/ACPI - Register Map

This register map is generated from the PeakRDL specification (`rtl/pm_acpi/peakrdl/pm_acpi_regs.rdl`) and matches the synthesized register block in `rtl/pm_acpi/pm_acpi_regs.sv`. Offsets are byte offsets from the block base address. All registers are 32 bits wide with 32-bit access.

Note: This block does not implement the classic ACPI PM1a/PM1b fixed-hardware layout (PM1_STS/PM1_EN/PM1_CNT at 0x00-0x0C, GPE0/GPE1). It is a custom PeakRDL register file. Fields such as BM_STS, GBL_STS, GBL_EN, SCI_EN, BM_RLD, GBL_RLS and a second GPE bank (GPE1) do not exist in the RTL.

3.1 Register Summary

Offset	Name	Access	Reset	Description
0x000	ACPI_CO NTROL	RW	0x00000000 0	Global control and power state
0x004	ACPI_STA TUS	W1C	0x00000000 0	Global status and power events
0x008	ACPI_INT _ENABLE	RW	0x00000000 0	Interrupt enable mask
0x00C	ACPI_INT _STATUS	W1C	0x00000000 0	Interrupt status
0x010	PM1_CO NTROL	RW	0x00000000 0	PM1 control (sleep, button override)
0x014	PM1_STA TUS	W1C	0x00000000 0	PM1 status flags
0x018	PM1_ENA BLE	RW	0x00000000 0	PM1 event enable mask
0x01C	(Reserve d)	-	-	Reserved
0x020	PM_TIME R_VALUE	RO	0x00000000 0	PM Timer current value (32-bit)
0x024	PM_TIME R_CONFI G	RW	0x00000001 B	PM Timer clock divider
0x028-0x02C	(Reserve d)	-	-	Reserved
0x030	GPE0_ST ATUS_LO	W1C	0x00000000 0	GPE0 status bits [15:0]
0x034	GPE0_ST	W1C	0x00000000	GPE0 status bits [31:16]

Offset	Name	Access	Reset	Description
	ATUS_HI		0	
0x038	GPE0_EN ABLE_LO	RW	0x00000000 0	GPE0 enable bits [15:0]
0x03C	GPE0_EN ABLE_HI	RW	0x00000000 0	GPE0 enable bits [31:16]
0x050	CLOCK_G ATE_CTR L	RW	0xFFFFFFFF F	Clock gating control [31:0]
0x054	CLOCK_G ATE_STA TUS	RO	0x00000000 0	Clock gate status
0x058	POWER_ DOMAIN _CTRL	RW	0x0000000F F	Power domain control [7:0]
0x05C	POWER_ DOMAIN _STATUS	RO	0x00000000 0	Power domain status
0x060	WAKE_S TATUS	W1C	0x00000000 0	Wake event sources
0x064	WAKE_E NABLE	RW	0x00000000 0	Wake event enable mask
0x068	RESET_C TRL	RW	0x00000000 0	Reset generation control
0x06C	RESET_S TATUS	RO	0x00000000 0	Reset source information

Access legend: RW = read/write, RO = read-only (hardware-updated), W1C = read status / write 1 to clear.

Addresses 0x070-0xFFF are decoded as reserved. The register block only decodes the offsets listed above; unmapped reads in the decoded range return 0.

3.2 ACPI_CONTROL (0x000)

Bits	Name	Access	Reset	Description
0	acpi_enable	RW	0	Enable ACPI functionality (0=disabled, 1=enabled)
1	pm_timer_enable	RW	0	Enable PM Timer (0=stopped, 1=running)
2	gpe_enable	RW	0	Enable GPE event processing
5:4	current_state	RO	0	Current power state (0=S0, 1=S1, 3=S3), hardware-updated
6	low_power_req	RW	0	Request low power mode entry
7	soft_reset	RW	0	Soft reset PM controller (write 1, auto-clears)
31:8	reserved	RO	0	Reserved

3.3 ACPI_STATUS (0x004)

Write 1 to clear each bit.

Bits	Name	Access	Reset	Description
0	pme_status	W1C	0	Power Management Event occurred
1	wake_status	W1C	0	System woke from low power state
2	timer_overflow	W1C	0	PM Timer overflow occurred
3	state_transition	W1C	0	Power state

Bits	Name	Access	Reset	Description
	tion			transition complete
31:4	reserved	RO	0	Reserved

3.4 ACPI_INT_ENABLE (0x008)

Bits	Name	Access	Reset	Description
0	pme_enable	RW	0	Enable interrupt on PME event
1	wake_enable	RW	0	Enable interrupt on wake event
2	timer_ovf_enable	RW	0	Enable interrupt on PM timer overflow
3	state_trans_enable	RW	0	Enable interrupt on power state transition
4	pm1_enable	RW	0	Enable interrupt on any PM1 event
5	gpe_intenable	RW	0	Enable interrupt on any GPE event
31:6	reserved	RO	0	Reserved

3.5 ACPI_INT_STATUS (0x00C)

Write 1 to clear each bit.

Bits	Name	Access	Reset	Description
0	pme_int	W1C	0	PME interrupt pending
1	wake_int	W1C	0	Wake interrupt pending
2	timer_ovf_int	W1C	0	Timer overflow

Bits	Name	Access	Reset	Description
				interrupt pending
3	state_trans_int	W1C	0	State transition interrupt pending
4	pm1_int	W1C	0	PM1 interrupt pending
5	gpe_int	W1C	0	GPE interrupt pending
31:6	reserved	RO	0	Reserved

3.6 PM1_CONTROL (0x010)

Bits	Name	Access	Reset	Description
2:0	sleep_type	RW	0	Sleep type (0=S0, 1=S1, 3=S3)
3	sleep_enable	RW	0	Enter sleep state (write 1; hardware auto-clears one cycle later)
4	pwrbtn_override	RW	0	Override power button behavior
5	slpbtn_override	RW	0	Override sleep button behavior
31:6	reserved	RO	0	Reserved

3.7 PM1_STATUS (0x014)

Write 1 to clear each bit.

Bits	Name	Access	Reset	Description
0	tmr_sts	W1C	0	PM Timer

Bits	Name	Access	Reset	Description
				carry/overflow
1	pwrbtn_sts	W1C	0	Power button pressed
2	slpbtn_sts	W1C	0	Sleep button pressed
3	rtc_sts	W1C	0	RTC alarm occurred
4	wak_sts	W1C	0	System wake event
31:5	reserved	RO	0	Reserved

3.8 PM1_ENABLE (0x018)

Bits	Name	Access	Reset	Description
0	tmr_en	RW	0	Enable PM timer events
1	pwrbtn_en	RW	0	Enable power button events
2	slpbtn_en	RW	0	Enable sleep button events
3	rtc_en	RW	0	Enable RTC alarm events
31:4	reserved	RO	0	Reserved

3.9 PM_TIMER_VALUE (0x020)

Bits	Name	Access	Reset	Description
31:0	timer_value	RO	0	Current 32-bit PM timer count (hardware-updated)

The counter is 32 bits and increments at a divided clock rate (see PM_TIMER_CONFIG). At the default divider it advances at a 3.579545 MHz equivalent rate and rolls over roughly every 1200 seconds.

3.10 PM_TIMER_CONFIG (0x024)

Bits	Name	Access	Reset	Description
15:0	timer_div	RW	0x001B	Clock divider: $\text{timer_clk} = \text{pm_clk} / (\text{timer_div} + 1)$
31:16	reserved	RO	0	Reserved

Reset value 0x001B (27 decimal) divides by 28, yielding the 3.579545 MHz equivalent tick for a ~100 MHz pm_clk.

3.11 GPE0_STATUS_LO (0x030)

Write 1 to clear. Covers GPE sources 0-15.

Bits	Name	Access	Reset	Description
15:0	gpe_status	W1C	0	General Purpose Event status bits 0-15
31:16	reserved	RO	0	Reserved

3.12 GPE0_STATUS_HI (0x034)

Write 1 to clear. Covers GPE sources 16-31.

Bits	Name	Access	Reset	Description
15:0	gpe_status	W1C	0	General Purpose Event status bits 16-31
31:16	reserved	RO	0	Reserved

3.13 GPE0_ENABLE_LO (0x038)

Covers GPE sources 0-15.

Bits	Name	Access	Reset	Description
15:0	gpe_enable	RW	0	General Purpose Event enable bits 0-15
31:16	reserved	RO	0	Reserved

3.14 GPE0_ENABLE_HI (0x03C)

Covers GPE sources 16-31.

Bits	Name	Access	Reset	Description
15:0	gpe_enable	RW	0	General Purpose Event enable bits 16-31
31:16	reserved	RO	0	Reserved

The 32 GPE sources are exposed as two 16-bit LO/HI register pairs. There is no second GPE bank (no GPE1).

3.15 CLOCK_GATE_CTRL (0x050)

Bits	Name	Access	Reset	Description
31:0	clk_gate_c	RW	0xFFFFFFFF	Clock gate enable per

Bits	Name	Access	Reset	Description
	trl			block (0=gated/off, 1=enabled/on)

3.16 CLOCK_GATE_STATUS (0x054)

Bits	Name	Access	Reset	Description
31:0	clk_gate_status	RO	0	Actual clock gate state per block (hardware-updated)

3.17 POWER_DOMAIN_CTRL (0x058)

Bits	Name	Access	Reset	Description
7:0	pwr_domain_ctrl	RW	0xFF	Power domain enable per domain (0=off, 1=on)
31:8	reserved	RO	0	Reserved

3.18 POWER_DOMAIN_STATUS (0x05C)

Bits	Name	Access	Reset	Description
7:0	pwr_domain_status	RO	0	Actual power domain state (hardware-updated)
31:8	reserved	RO	0	Reserved

3.19 WAKE_STATUS (0x060)

Write 1 to clear each bit.

Bits	Name	Access	Reset	Description
0	gpe_wake	W1C	0	Woke from GPE event
1	pwrbtn_wake	W1C	0	Woke from power

Bits	Name	Access	Reset	Description
				button
2	rtc_wake	W1C	0	Woke from RTC alarm
3	ext_wake	W1C	0	Woke from external signal
31:4	reserved	RO	0	Reserved

3.20 WAKE_ENABLE (0x064)

Bits	Name	Access	Reset	Description
0	gpe_wake_en	RW	0	Enable wake from GPE events
1	pwrbtn_wake_en	RW	0	Enable wake from power button
2	rtc_wake_en	RW	0	Enable wake from RTC alarm
3	ext_wake_en	RW	0	Enable wake from external signal
31:4	reserved	RO	0	Reserved

3.21 RESET_CTRL (0x068)

Bits	Name	Access	Reset	Description
0	sys_reset	RW	0	Generate system reset (write 1, auto-clears)
1	periph_re	RW	0	Generate peripheral

Bits	Name	Access	Reset	Description
	set			reset (write 1, auto-clears)
31:2	reserved	RO	0	Reserved

3.22 RESET_STATUS (0x06C)

Bits	Name	Access	Reset	Description
0	por_reset	RO	0	Last reset was power-on reset
1	wdt_reset	RO	0	Last reset was watchdog timeout
2	sw_reset	RO	0	Last reset was software initiated
3	ext_reset	RO	0	Last reset was external pin
31:4	reserved	RO	0	Reserved

3.23 Implementation Notes (known RTL deviations)

The register file decodes and stores these registers, but several software-visible behaviors are not yet wired through to the PM core. These are RTL issues tracked separately (not fixed in documentation); they are documented here so software does not rely on behavior the current RTL does not provide:

- ACPI_CONTROL.soft_reset (bit 7) and RESET_CTRL.sys_reset/periph_reset are not connected to the core; the reset-request outputs are hardwired to 0. Writing them auto-clears but has no effect.

- ACPI_CONTROL.low_power_req (bit 6) and PM1_CONTROL.pwrbtn_ovr/slpbtn_ovr (bits 4/5) are decoded but unused by the core.
 - PM1_ENABLE per-source enables (tmr_en/pwrbtn_en/slpbtn_en/rtc_en) are decoded but do not currently gate PM1 status or the PM1 interrupt.
 - Clearing GPE0_STATUS_LO/HI clears the register bits, but the core's internal GPE sticky status has no clear path, so the aggregated GPE interrupt source can remain asserted until reset.
 - W1C status fields (ACPI_STATUS, ACPI_INT_STATUS, PM1_STATUS, WAKE_STATUS) have a hardware-update path whose next input is undriven in the current build; treat set-and-hold behavior of these status bits as unreliable until fixed.
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